

Sergii Tsymbaliuk

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Summary

Senior software engineer specializing in large-scale distributed systems, cloud infrastructure, and data platforms — with high-impact delivery at LinkedIn, AWS, Roku, Apple, and Google. Built petabyte-scale HDFS storage systems at LinkedIn, improved IAM reliability for billions of AWS customers, and drove ~15% engagement growth at Roku through ML personalization — now advancing infrastructure automation and operational intelligence with AI-native tooling.

Skills

Languages: Java, Scala, Python, TypeScript
Infrastructure & Cloud: Kubernetes, Docker, Linux, Shell Scripting, AWS, Google Cloud Platform
Data Platforms: Hadoop/HDFS, Apache Spark, Apache Airflow, Apache Kafka, Apache Gobblin
Storage: MySQL, DynamoDB, Apache Cassandra, MongoDB, Redis
APIs & Protocols: REST, gRPC, HTTP/HTTPS, JSON, Protobuf
Domains: Distributed Systems, Large-Scale Systems, Microservices, Big Data, AI/ML, Cloud Infrastructure
Practices: CI/CD, Platform Engineering, Static Code Analysis, Network Architecture, Agile/Scrum
AI Tooling: Claude Code, LLM, MCP

Experience

Senior Software Engineer — LinkedIn

Jun 2024 – Present | Sunnyvale, CA

Project: Data Lake Management (DLM) within Core Infrastructure's Analytics & AI Storage team — an intelligent storage management ecosystem spanning Quota Management & Enforcement, HDFS Capacity Planning & Forecasting, Hadoop Corpus Reduction (projected \$30M+ savings in H2-FY26), and Space Mover for automated data migration across HDFS clusters.

Achievements:

- Owned Space Management Service launch: automated HDFS mount-point lifecycle (provisioning, encryption zones, federation publishing) and cutover orchestration; established SLOs, runbooks, and

monitoring for production-grade reliability.

- Led build-out of the Quota Management Service: designed enforcement-group model, implemented mid-tier hard enforcement and throttling, and delivered ETL pipelines for storage visibility and dataset attribution.
- Drove platform observability: migrated quota dashboards to real-time service data, automated dataset-to-enforcement-group mappings via daily ETLs, and extended the usage dashboard with top-path breakdown by storage and object count — improving accuracy and reducing manual operational overhead.
- Built an agent toolkit and skills library to automate quota and utilization analysis across core infrastructure, accelerating team workflows.

Toolset: Java, Scala, Python, Kubernetes, Docker, Linux, MySQL, Spring Framework, Hibernate, Hadoop, HDFS, Apache Airflow, Apache Spark, Apache Gobblin, [Openhouse](#), [Datahub](#), Claude Code, MCP

Senior Software Engineer — Amazon AWS

Mar 2022 – Jun 2024 / San Jose, CA

Project: AWS IAM — a foundational AWS service with billions of internal and external customers, requiring extreme availability, strict transactional consistency, and low latency across all global regions.

Achievements:

- Delivered a data retention solution for closed AWS accounts: led design, implementation, and multi-year historical data backfill. Explicitly removed PII to satisfy GDPR and CCPA requirements, mitigated legal and reputational risk, and eliminated manual Customer Support retention procedures.
- Implemented a block-listing solution to prevent publicly disclosed account IDs from appearing in documentation and code examples, closing a security and compliance exposure at scale.
- Automated infrastructure and service configuration generation for new AWS regions, reducing time-to-launch from weeks to hours and eliminating manual effort and human error.

Toolset: Java, SpringBoot, Guice, Hibernate, MySQL, Ruby on Rails, TypeScript, Docker, Linux, AWS: EC2, ECS, CloudFormation, DynamoDB, Kinesis, S3, SQS, SNS, ELB

Senior Software Engineer — Roku

Aug 2020 – Mar 2022 / San Jose, CA

Project: Roku Search and Recommendations — a suite of microservices and data pipelines powering content search and personalized recommendations across the Roku platform and streaming channels.

Achievements:

- Designed and launched ML-backed personalized content collections aggregating titles across user subscriptions based on viewing history and preferences, driving ~15% improvement in user engagement.
- Built a probabilistic cardinality metrics framework using HyperLogLog, enabling multi-dimensional customer engagement measurement at scale.
- Root-caused and resolved a critical news feed latency bottleneck, reducing response time from 10 seconds to 200 milliseconds under peak load.

Toolset: Java, Spring Framework, SparkJava, Guice, Apache Kafka, Redis, Apache Flume, Docker, EnvoyProxy, gRPC, AWS: DynamoDB, S3, ECS, EKS, ELB, CloudFormation

Senior Software Engineer — Apple

Jul 2019 – Aug 2020 / Cupertino, CA

Project: Apple Media Products — a distributed transcoding system at scale used by Apple TV+, iTunes, and the App Store to convert production media assets into consumer formats. Contributed to the architectural shift from a legacy bare-metal media-conversion system to an elastic cloud-native solution.

Achievements:

- Integrated the media processing task scheduler with the media-encoding system for Apple TV+, enabling elastic workload distribution across the cloud-native fleet.
- Built a pub/sub notification microservice for tracking media transcoding status and artifact readiness, enabling downstream consumers to react to pipeline events without polling.
- Implemented a cluster telemetry service to collect hardware capabilities, capacity, and utilization data from Kubernetes nodes, supporting resource optimization across the fleet.
- Standardized code quality gates by introducing JaCoCo coverage enforcement and SonarQube Quality Gates into CI, with automated PR decoration for inline review feedback.
- Mentored junior engineers and contributed to engineering hiring.

Toolset: Java, Jetty, Guice, gRPC, Docker, Kubernetes, fabric8, AWS SDK, Apache Kafka, Apache Cassandra, Redis, Apache ZooKeeper, Apache Curator, AWS DynamoDB

Software Engineer — EPAM for Google (YouTube Creator Academy)

Apr 2018 – Jul 2019 / Mountain View, CA

Project: Online learning platform for YouTube content creators; backend engineering.

Achievements:

- Implemented content popularity and user activity metrics, giving creators visibility into engagement and reach.
- Integrated Google's ML-based recommendation service to surface more relevant courses and improve user engagement.
- Automated the Creator Awards ETL pipeline, eliminating manual data processing and freeing 16 developer-hours per month.
- Migrated the App Engine service to Java 8 and Google Cloud Endpoints v2.0, enabling gRPC support for downstream services.

Toolset: Java, Python, TypeScript, AngularJS, REST, Protobuf, gRPC, FlumeJava, MapReduce, Apache Beam, Cloud Dataflow, BigTable, BigQuery, Cloud Datastore, MySQL, Bazel, Google Cloud Platform

Software Engineer — EPAM for Google (Google Search Appliance)

Apr 2016 – Apr 2018 / Mountain View, CA

Project: Google-manufactured enterprise search appliance for corporate intranet and file systems; platform infrastructure team.

Achievements:

- Refactored the build automation service from synchronous to async job execution, cutting average build time from 2 hours to 30 minutes.
- Designed and implemented a custom VPN solution (OpenVPN) for connecting developer machines to remote devices in isolated networks, reducing debug setup time from 2 hours to minutes.
- Built a multi-layer RAID and filesystem health-check module for the self-monitoring and recovery service, specifically targeting device failures caused by unsuccessful patch and upgrade attempts — reducing bricked devices and eliminating the need to ship hardware back to Google for recovery.
- Created a distributed log collection system for a bare-metal test machine pool, enabling teams to aggregate installation and debug logs across remote infrastructure.

Toolset: Java, Python, Django, SQLAlchemy, REST, Protobuf, gRPC, Google Borg, Dremel, BigTable, Colossus, MySQL, Bazel, Linux, Docker, Shell Scripting, iptables, OpenVPN, Google Cloud Platform

Application Support Team Lead — EPAM for Barclays

Aug 2012 – Apr 2016 / Kyiv, Ukraine

Project: Application support and incident management for Barclays Capital banking systems, leading a distributed team across multiple time zones.

Achievements:

- Played a pivotal role in migrating a core banking system to a microservices architecture: contributed to architectural design, planning, and implementation to enhance system scalability and operational efficiency.
- Drove Agile Scrum and Kanban adoption across the support team, achieving ~30% increase in team throughput and 60% reduction in average incident response time.

Toolset: Java, MongoDB, Linux, Microsoft SQL Server, T-SQL, C#/.NET, Autosys, Jira, Perforce, ServiceNow, Agile/Scrum/Kanban

Product Manager — ING Bank Ukraine

May 2011 — Aug 2012 / Kyiv, Ukraine

- Led the release of a Cyrillic transliteration module for the SWIFT processing system, shifting international transfers from paper-based to electronic SWIFT transactions and significantly reducing processing time.

Toolset: SWIFT, Oracle Application Server, Oracle Database, Java, JavaScript

Senior Infrastructure Engineer & Network Architect — ING Bank Ukraine

Nov 2005 — May 2011 / Kyiv, Ukraine

- Designed and deployed highly available internet and WAN connectivity across multiple bank data centers.
- Migrated the network core to a split architecture with VRRP, BGP, OSPF, and IPSec-based load balancing, eliminating single points of failure.
- Introduced centralized AAA on network devices using LDAP, SNMP, and Cisco IOS; automated configuration management and backup.
- Deployed a NetFlow-based traffic monitoring and analysis platform across data centers.
- Established multipath LAN/WAN routing using BGP/OSPF with route redistribution, eliminating cold-standby infrastructure dependency.

Toolset: Cisco IOS, Checkpoint Firewall, Linux, MS SQL Server, Microsoft Active Directory, TCP/IP, DNS, OSPF, BGP, IPSec, VPN, VMware ESXi, Citrix XenApp

Lead System Administrator — Naftogaz of Ukraine DC Gas-Heat

Sep 2003 — Oct 2005 / Kyiv, Ukraine

Achievements:

- Built the company's IT infrastructure from the ground up, including servers, networking, and core services.

Toolset: Linux, Windows Server, TCP/IP, Network Architecture, Shell Scripting, Automation

Education

- **M.S. in Computer Science (With Honors)**, 2004 — National Technical University of Ukraine "Kyiv Polytechnic Institute" *Specialization: Software of Automated Systems*. Thesis: Developed a multi-agent information system using neural networks and artificial life evolution algorithms.
- **B.S. in Computer Science**, 2002 — National Technical University of Ukraine "Kyiv Polytechnic Institute"
- **M.S. in Finances (Economics & Enterprise)**, 2009 — Kyiv National Economic University named after Vadym Hetman *Specialization: Financial Management in the Business Sphere*

Publications

- **Authentication Authorization Accounting: Integrating Cisco with Microsoft Active Directory** — *System Administrator Magazine*, N°1-2 (98-99), 2011 | samag.ru/archive/more/103

Certifications

- Google Cloud Certified Professional — 2018
- ITIL v3 Foundation — 2013
- Cisco Certified Network Professional (CCNP) — 2008
- Cisco Certified Network Associate (CCNA) — 2005